

NAMIBIA UNIVERSITY OF SCIENCE AND TECHNOLOGY

Faculty of Computing and Informatics

PHASE 2 DELIVERABLE

Software Verification and Validation

SVV711S

Student Name	Student Number	Contribution
Selma Iileka	223102415	100%
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Date of Submission: 27 April 2026

1. TIMELINE OF ACTIVITIES (Phase 2)

Date	Activity	Participants
01/03	Maria and Mukowa scheduled meeting with Mr. Kasika	Maria, Mukowa
05/03	Virtual meeting with Mr. Kasika (Google Meet)	Maria, Mukowa
05/03	Mr. Kasika sent Mood Tracker documentation	Mr. Kasika
05/03	Mr. Kasika instructed to use "version1" branch on GitHub	Mr. Kasika
24/03	Swapping list released – Selma's name included	Selma
24/03	Josia Taapopi emailed Dr Kapuire admitting attendance mistake	Josia
25/03	Eben Erasmus promised to intervene for Selma	Selma, Eben
08/04	Swapping list released – Selma swapped with Maria	Selma, Maria
08/04	Selma emailed Dr Kapuire requesting attendance fix	Selma
10/04	Eben Erasmus confirmed Selma's 100% attendance	Eben
21/04	Selma spoke face to face with Dr Kapuire	Selma
24/04	Selma emailed Maria introducing as new partner	Selma, Maria
24/04	Maria sent Mood Tracker document	Maria
24-25/04	Reviewed the Mood Tracker system and code	Selma, Maria
25/04	Online meeting between Selma and Maria	Selma, Maria
25-27/04	Phase 2 document compilation	Selma, Maria

2. SYSTEM EVALUATION

System Name: Odyssey (Mood Tracker Mobile Application)

Client: Mr. Keith Kasika

Platform: Flutter (Android, iOS)

Repository Branch: version1

2.1 What the System Contains (Based on Code Analysis)

The source code was analyzed. The following components were observed:

Core Features:

- Onboarding flow with student profile collection (name, field of study, year, mode of study, consent)
- Mood logging with date/time picker, emoji mood selector, and 3-tier emotion chip selector
- Activity tagging on mood entries (categories: Exercise, Social, Work, etc.)
- Mood log list with grouped records, edit and delete per entry
- Analytics screen with bar chart (mood count) and line chart (mood variation over time)
- Google Sheets sync – exports mood data with user ID, incremental and full sync
- Auto-sync on connectivity change with offline queue
- Push notifications – 3 configurable daily reminder times (morning, afternoon, evening)
- Settings screen – profile view, notification time pickers, manual Google Sheets sync
- Terms and Conditions screen (POPIA compliance, NUST research context)

- Local offline storage via Hive database
- Responsive layout support

2.2 Emotion Wheel Implementation

A 3-tier hierarchical emotion system implemented as chip selectors, not a radial wheel.

- File: lib/src/features/mood_records/domain/emotion_wheel/emotion_wheel.dart
- 6 primary emotions: Happy, Sad, Angry, Fearful, Bad, Disgust – each color-coded
- Secondary emotions: 33 total (e.g. Lonely, Anxious, Frustrated)
- Tertiary emotions: Approximately 72 total (e.g. Hopeless, Withdrawn, Nauseated)
- UI implementation: add_mood_record_form.dart renders them as ChoiceChip and FilterChip widgets in cascading rows (Primary to Secondary to Tertiary)

A separate legacy mood picker also exists in mood_configurations.dart with 5 emoji moods (Great, Positive, Alright, Bad, Awful) using SVG icons. Both systems coexist in the code.

2.3 AI Prediction Logic

Not found in the code. All .dart files were searched for terms including AI, machine learning, tensorflow, predict, classif, neural, and model. No matches related to machine learning were found. The application operates on manual entry and rule-based logic only.

2.4 Mood Trigger Detection

Not found in the code. A search for trigger, threshold, alert, and pattern detection returned no relevant results. The only references found were:

- auto_sync_service.dart line 47: "New mood record detected" – this refers to a sync event, not mood analysis
- terms_screen.dart line 21: Mention of "Emotional and behavioral patterns" in consent text – legal language only, no implementation

2.5 Data Retention and Deletion Policy

Partial implementation was observed.

What exists:

- mood_record_repository.dart lines 102-112: deleteMoodRecord(int key) – deletes individual records from Hive local storage
- student_info_repository.dart lines 38-40: deleteStudentInfo() – deletes profile from Hive
- student_info_viewer.dart lines 129-172: UI confirmation dialog for deleting student profile
- terms_screen.dart references POPIA (Namibian data protection law) and states users can request data deletion

What does not exist:

- No automated data expiry or retention period
- No deletion from Google Sheets when local record is deleted (dialog text states: "This will not delete data from Google Sheets")
- No data anonymization implementation

3. OBSERVED ISSUES (Bugs and Code Quality)

The following issues were identified during code analysis and are documented as observed facts.

3.1 Critical Issues

ID	Issue	Location
B01	router variable used before it is defined – will cause runtime crash	app_router.dart, line 149
B02	Emotion map key collision: 'Inferior' appears under both 'Depressed' and 'Insecure' – only one survives	emotion_wheel.dart, lines 197, 297, 557-560
B03	/terms route is commented out but Settings screen links to it – will cause navigation crash	app_router.dart lines 104-107, settings_screen.dart line ~140
B04	/profile-info route is never registered but Settings screen links to it – will cause crash	settings_screen.dart line ~92

3.2 Moderate Issues

ID	Issue	Location
B05	Notification scheduling is commented out – users can set times but notifications never send	settings_screen.dart
B06	MoodRecord.fromJson is commented out – records cannot be deserialized from JSON	mood_record.dart line ~18
B07	Incremental sync assumes new records are always at end of Hive box – may sync wrong records if order changes	auto_sync_service.dart
B08	Delete from device does not delete from Google Sheets – privacy gap	student_info_viewer.dart line 150

3.3 Minor Issues (Code Quality)

ID	Issue	Location
B09	Multiple duplicate files exist (onboarding_screen, student_info_viewer, gsheets_test, settings_screen)	Multiple directories
B10	Embarrassed appears under both Disapproving and Hurt – only one version retained in map	emotion_wheel.dart
B11	MoodConfiguration TODO left unresolved – two mood systems both wired to save path	mood_configurations.dart line 1

4. TECHNOLOGIES OBSERVED

Category	Technology
Framework	Flutter (Dart)
Target platforms	Android, iOS
State management	Riverpod (flutter_riverpod ^2.3.5)
Navigation	GoRouter (go_router ^7.1.1)
Local database	Hive + Hive Flutter (hive ^2.2.3)

Category	Technology
Code generation	Freezed, build_runner, hive_generator, json_serializable, flutter_gen_runner
Charts	fl_chart (0.71.0)
Cloud sync	Google Sheets API via gsheets ^0.5.0
Connectivity	connectivity_plus ^7.0.0
Notifications	flutter_local_notifications ^19.4.2
Timezones	timezone ^0.10.1
Preferences	shared_preferences ^2.5.3
SVG icons	flutter_svg ^2.0.5 (24 custom mood SVG icons in /assets/mood_icons/)
UUIDs	uuid ^4.5.1
Localization	Flutter flutter_localizations + .arb files
UI extras	timeline_tile ^2.0.0, dynamic_color ^1.6.3
App name	odyssey
Package	com.example.odyssey_2
Research context	Built for NUST academic study

5. CLIENT COMMUNICATION (Maria's Phase 1)

The following communications occurred between Maria (with previous partner Mukowa) and client Mr. Kasika:

Date	Communication	Outcome
01/03	Email scheduling meeting	Client agreed to Thursday 13:00
05/03	Virtual meeting (Google Meet)	System explained, documentation received
05/03	Client email	Documentation attached, "version1" branch specified
05/03	GitHub usernames provided	donnaywillemse22@gmail.com, Mukowa8

Figure A1: Meeting scheduling email (1 March)

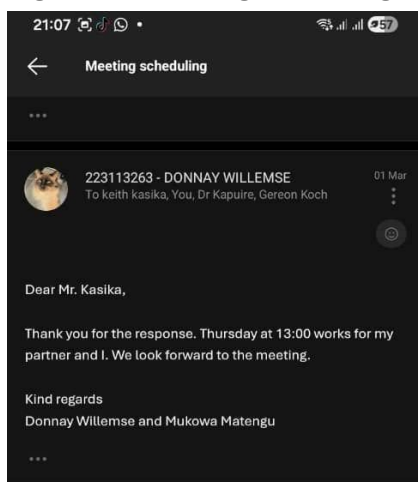


Figure A1: Email confirming Thursday 13:00 meeting with Mr. Kasika

Figure A2: GitHub usernames email

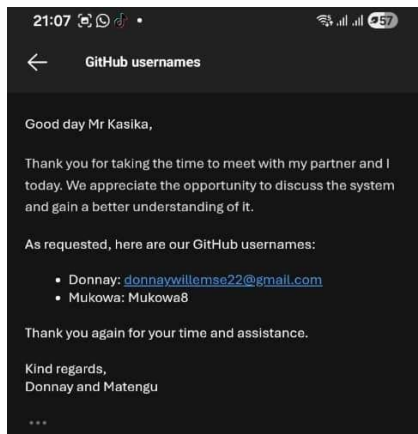


Figure A2: Email providing GitHub usernames to Mr. Kasika after the meeting

Figure A3: Client email with documentation and version1 instruction (8 March)

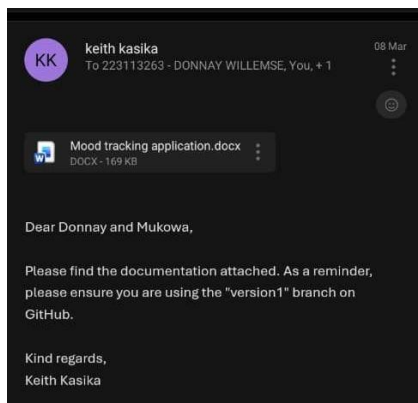


Figure A3: Mr. Kasika's email attaching documentation and specifying the version1 branch

6. PEER COLLABORATION

Date	Peer	Purpose	Outcome
24/03	Josia Taapopi (email)	Admitted attendance mistake	Acknowledged error
08/04	Dr Kapuire (email)	Requested attendance fix and fair swap	Advised to include class reps
10/04	Erastus Mukuve (email)	Confirmed Selma's 100% attendance	Sent list to Dr Kapuire
10/04	Eben Erasmus (email)	Explained attendance fault was class rep error	Confirmed Selma attended all classes
21/04	Dr Kapuire (face to face)	Asked for resolution	Told to email new partner
24/04	Maria Armas (WhatsApp)	Introduced as new partner	She agreed to meet and explain system
25/04	Maria Armas (online meeting)	System explanation and code review	Selma understood system requirements

Figure A4: Selma's WhatsApp message to Maria (24 April)

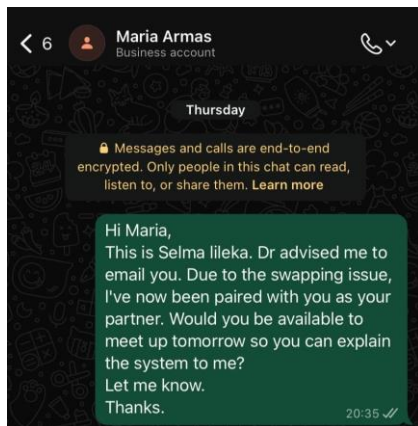


Figure A4: Selma contacting Maria on WhatsApp to arrange a meeting

Figure A5: Maria's WhatsApp response

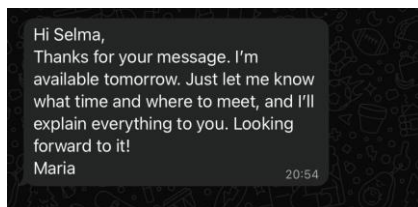


Figure A5: Maria confirming availability to meet and explain the system

7. EVIDENCE LOG

ID	Evidence	Date	Location
E01	Maria email – meeting scheduling	01/03	Appendix A, Figure A1
E02	Client response with meeting confirmation	01/03	Appendix A, Figure A2
E03	Client email with documentation and "version1"	05/03	Appendix A, Figure A3
E04	GitHub usernames email	05/03	Appendix A, Figure A4
E05	Selma email to Dr Kapuire (8 April)	08/04	Appendix A, Figure A5
E06	Eben Erasmus confirmation email	10/04	Appendix A, Figure A6
E07	Selma WhatsApp to Maria (24 April)	24/04	Appendix A, Figure A7
E08	Maria WhatsApp response	24/04	Appendix A, Figure A8
E09	Mood Tracker documentation	05/03	Appendix B
E10	Flutter code analysis findings	25/04	Appendix C

Appendix A: Email and Communication Evidence

Figure A6: Selma's email to Dr Kapuire (8 April)

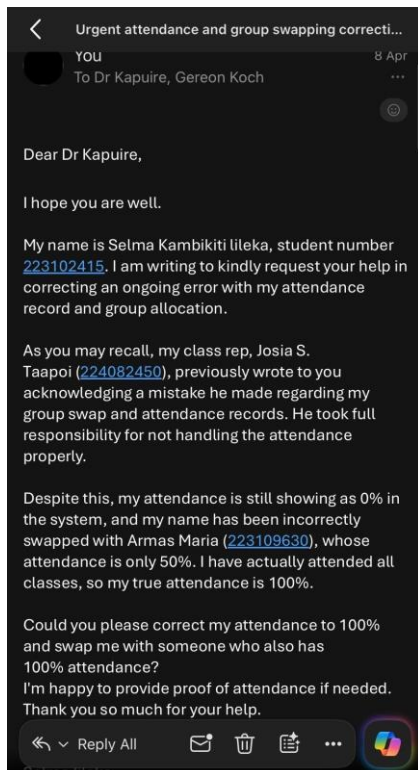


Figure A6: Selma's email to Dr Kapuire requesting attendance correction and group swap

Figure A7: Eben Erasmus confirmation email

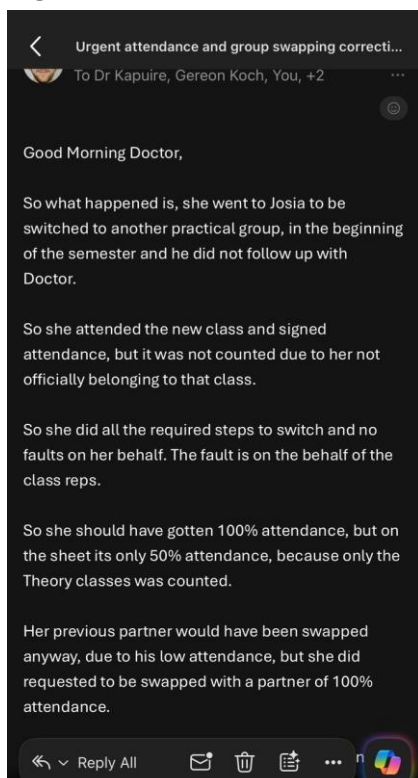


Figure A7: Email confirming Selma's 100% attendance and explaining the class rep error

Appendix C: Code Analysis Evidence

The following source files were manually reviewed during the code analysis:

- lib/src/features/mood_records/domain/emotion_wheel/emotion_wheel.dart
- lib/src/features/mood_records/presentation/add_mood_record_form.dart
- lib/src/Connectivity/auto_sync_service.dart
- lib/src/features/mood_records/data/mood_log/mood_record_repository.dart
- lib/src/features/profile/presentation/settings_screen.dart
- lib/src/routing/app_router.dart
- lib/src/features/profile/presentation/student_info_viewer.dart

Key findings from the manual review are documented in Sections 2 and 3 of this report.

End of Document